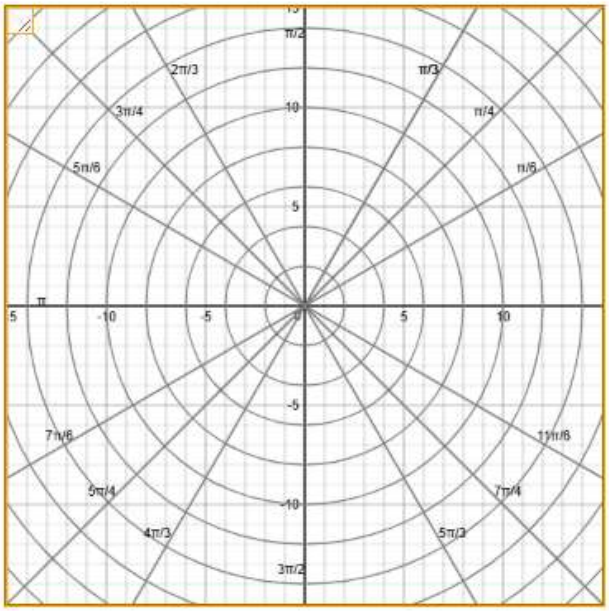
Calculus II Lesson 31B: Rectangular Equation and Polar Equation

1) Rectangular Equation: $x^2 + y^2 = 25$.

a) Write Corresponding Polar Equation: ___?___

b) Graph the polar equation.



3) Rectangular Equation: $x^2 = 5y + 2$.

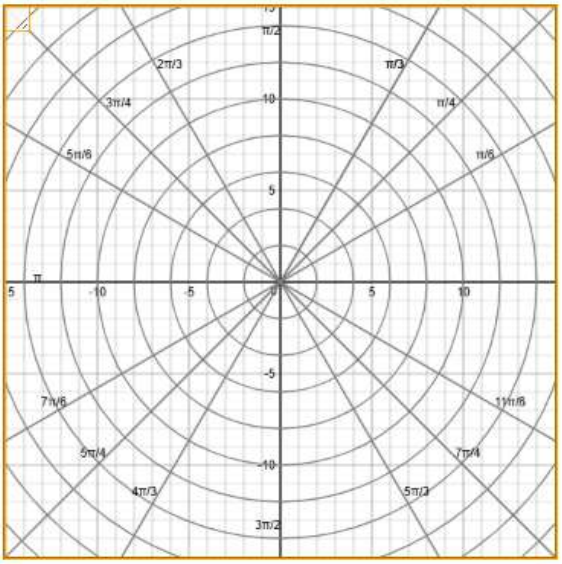
a) Write Corresponding Polar Equation: ___?___

(multiple choice)

A) $r^2 \cos^2\theta = 5r\sin\theta + 2$ B) $r\cos\theta = 5r\sin\theta + 2$

C) $\cos^2\theta = 5r\sin\theta + 2$ D) $r^2\cos^2\theta = 5r + 2$

b) Graph the polar equation.



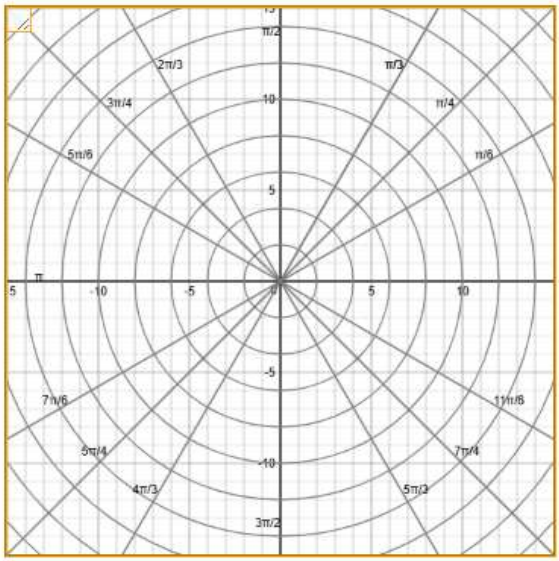
2) Rectangular Equation: $x = 8$.

a) Write Corresponding Polar Equation: ___?___

(multiple choice)

A) $r = \dfrac{8}{\sin\theta}$ B) $r = 8$ C) $r = \dfrac{8}{\cos\theta}$ D) $r = \dfrac{2}{\cos\theta}$

b) Graph the polar equation.



4) Polar Equation: $r = 6$.

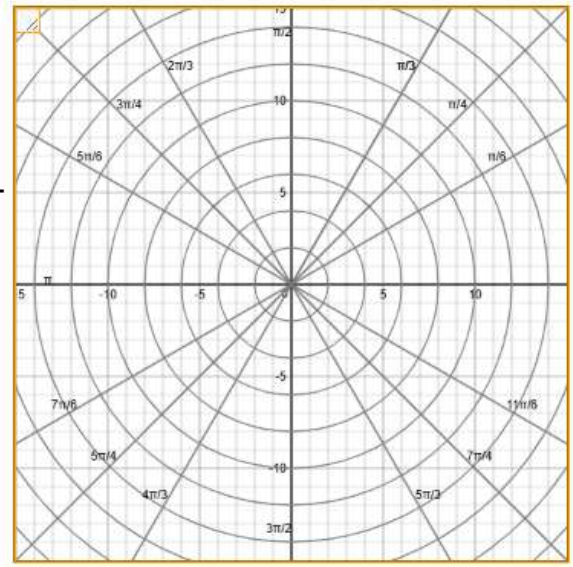
Hint: Square both sides first; $r^2 = 36$.

a) Write Corresponding Rectangular Equation: ___?___
(multiple choice)

A) $x^2 + y^2 = 6$ B) $x + y = 36$

C) $x^2 + y^2 = 36$ D) $x^2 + y^2 = 16$

b) Graph the polar equation.



5) Polar Equation: $r = 2\theta$.

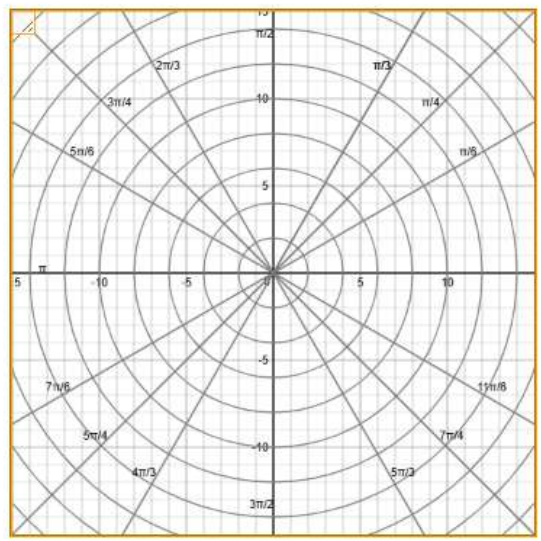
Hint: Square both sides of equation first; and use $\theta = \tan^{-1}\frac{y}{x}$.

a) Write Corresponding Rectangular Equation: ___?___

A) $x^2 + y^2 = 4\left(\frac{y}{x}\right)^2$ B) $x + y = 4\left(\tan^{-1}\frac{y}{x}\right)^2$

C) $x^2 + y^2 = 4\left(\tan^{-1}\frac{y}{x}\right)^2$ D) $x^2 + y^2 = \left(\tan^{-1}\frac{y}{x}\right)^2$

b) Graph the polar equation.



6) Polar Equation: $r = 5\cos\theta$.

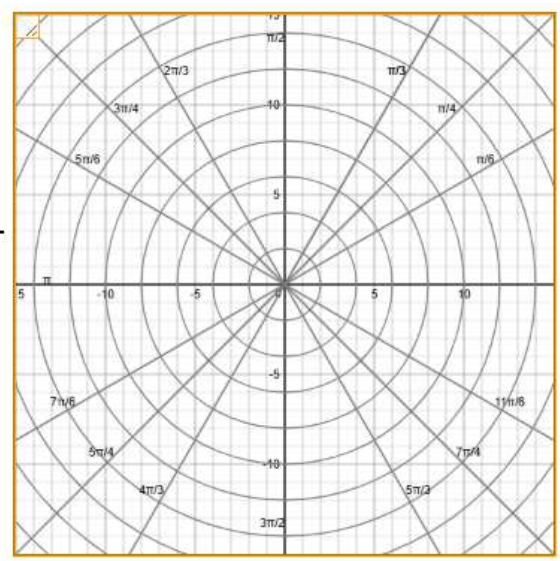
Hint: Multiply both sides by r; $r^2 = 5r\cos\theta$.

a) Write Corresponding Rectangular Equation: ___?___

A) $x^2 + y^2 = 5$ B) $x + y = 5x$

C) $x^2 + y^2 = 5x$ D) $x^2 + y = x$

b) Graph the polar equation.



7) Polar Equation: $r = 4 - 5\cos\theta$.

Hint: Multiply each side by r first; $r^2 = 4r - 5r\cos\theta$.

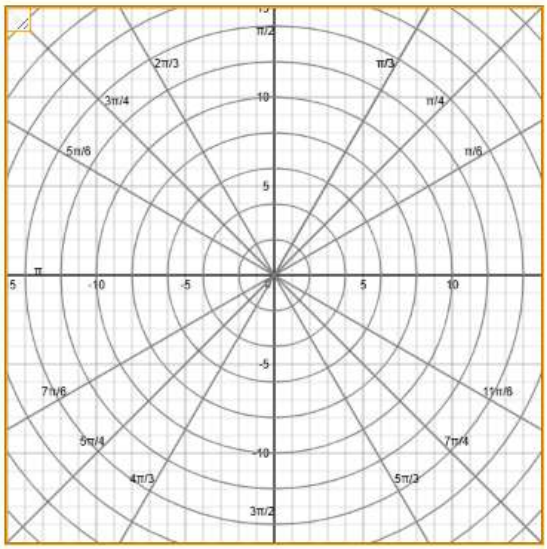
a) Write Corresponding Rectangular Equation: ?

(multiple choice)

A) $x^2 + y^2 = 4 - 5x$ B) $x^2 + y^2 = 4\left(\pm\sqrt{x^2 + y^2}\right) - 5x$

C) $x^2 + y^2 = \pm\sqrt{x^2 + y^2} - 5$ D) $x^2 + y^2 = 4x^2 - 5x$

b) Graph the polar equation.

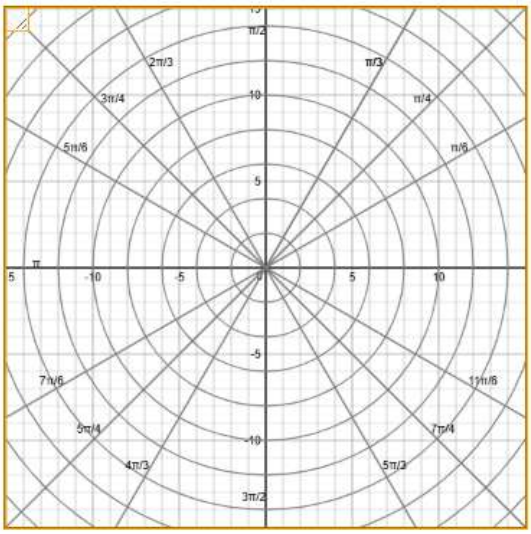


8) Polar Equation: $r = \dfrac{4}{1 + 3\cos\theta}$.

Hint: Multiply each side by $1+3\cos\theta$ first; $r + 3r\cos\theta = 4$.

a) Write Corresponding Rectangular Equation: ?

A) $\pm\sqrt{x^2 + y^2} + x = 4$ B) $\pm\sqrt{x^2 + y^2} + 3x = 4$

C) $\pm\sqrt{x^2 + y^2} + 3 = 4$ D) $\pm\sqrt{x^2 + y^2} + 3x = -4$

b) Graph the polar equation.